SUPPLEMENTARY INFORMATION

Nested Association Mapping (NAM) in quinoa (*Chenopodium quinoa*) for yield-related traits in the Peruvian Highlands

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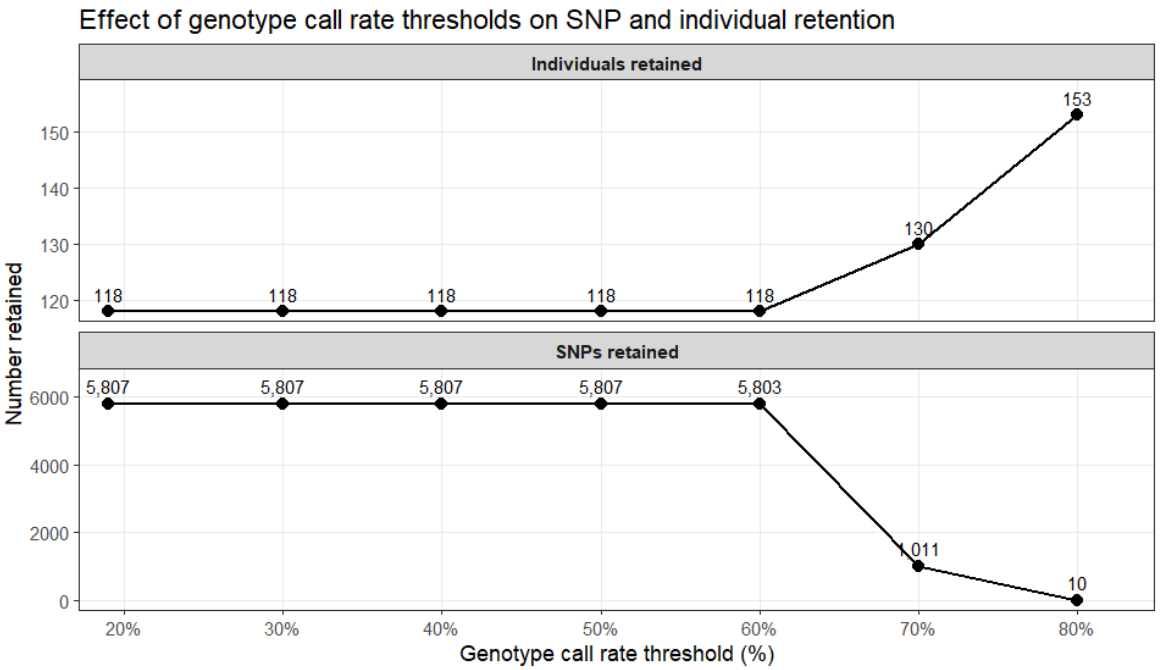
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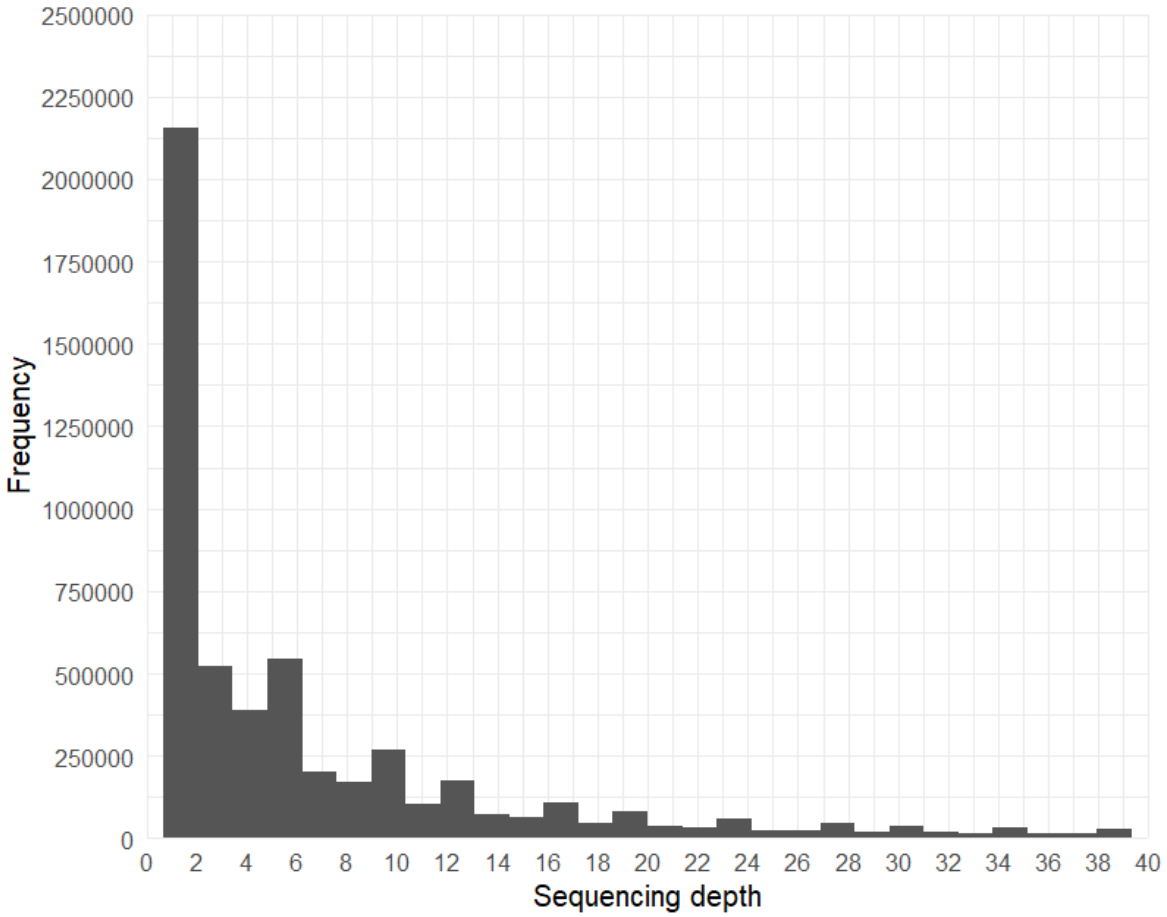
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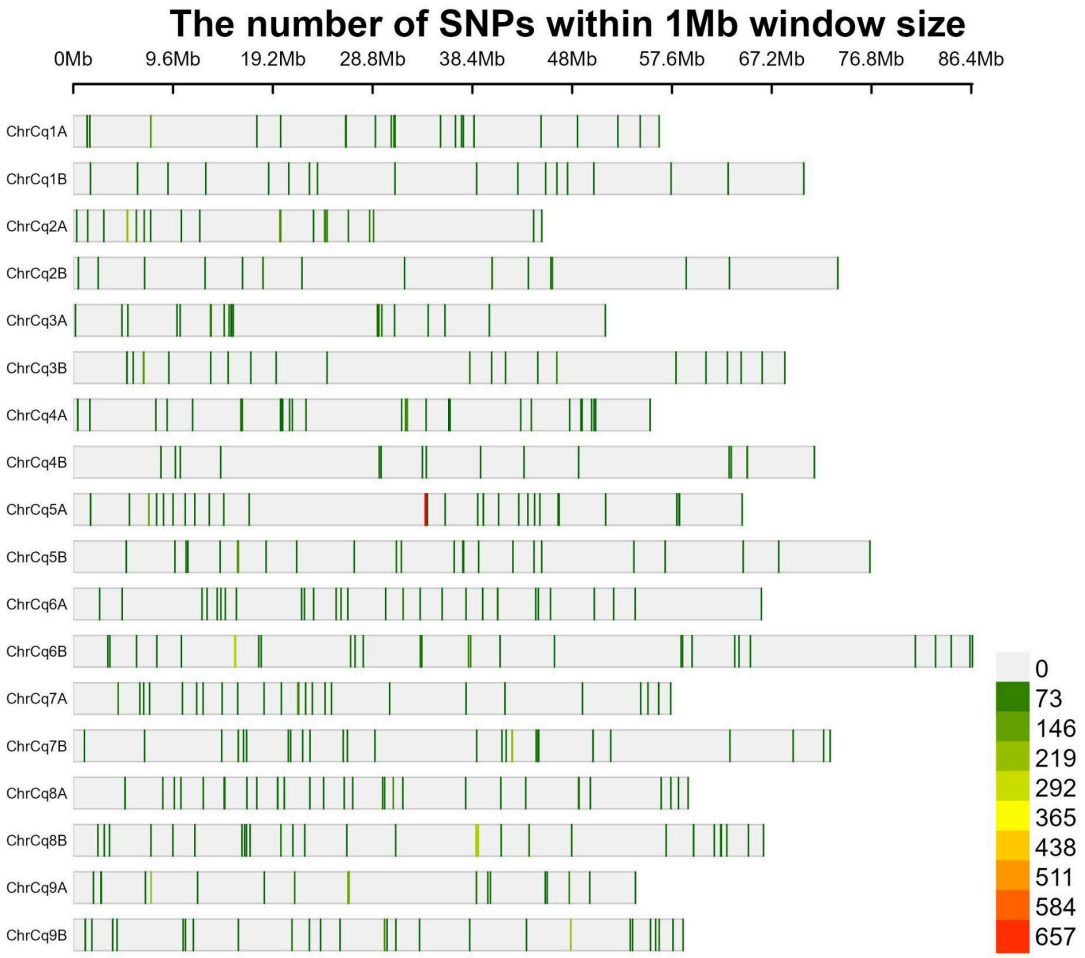
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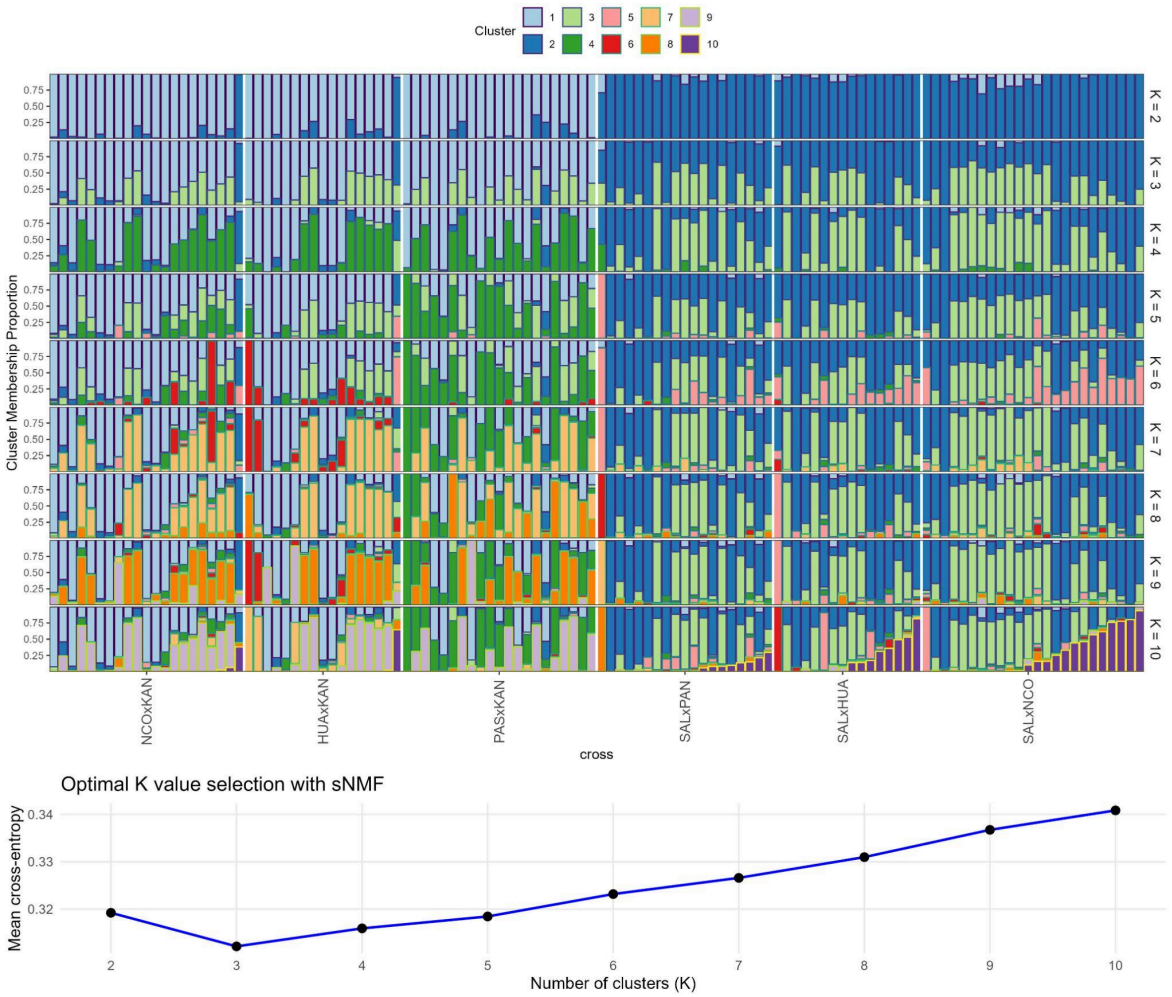
Supplementary Figure 1. Sensitivity of SNP and individual retention to minimum genotype call rate thresholds. Numbers of SNP markers retained after BCFtools and snpR filtering and individuals retained after genotype quality control across minimum genotype call rate thresholds ranging from 20% to 80%.



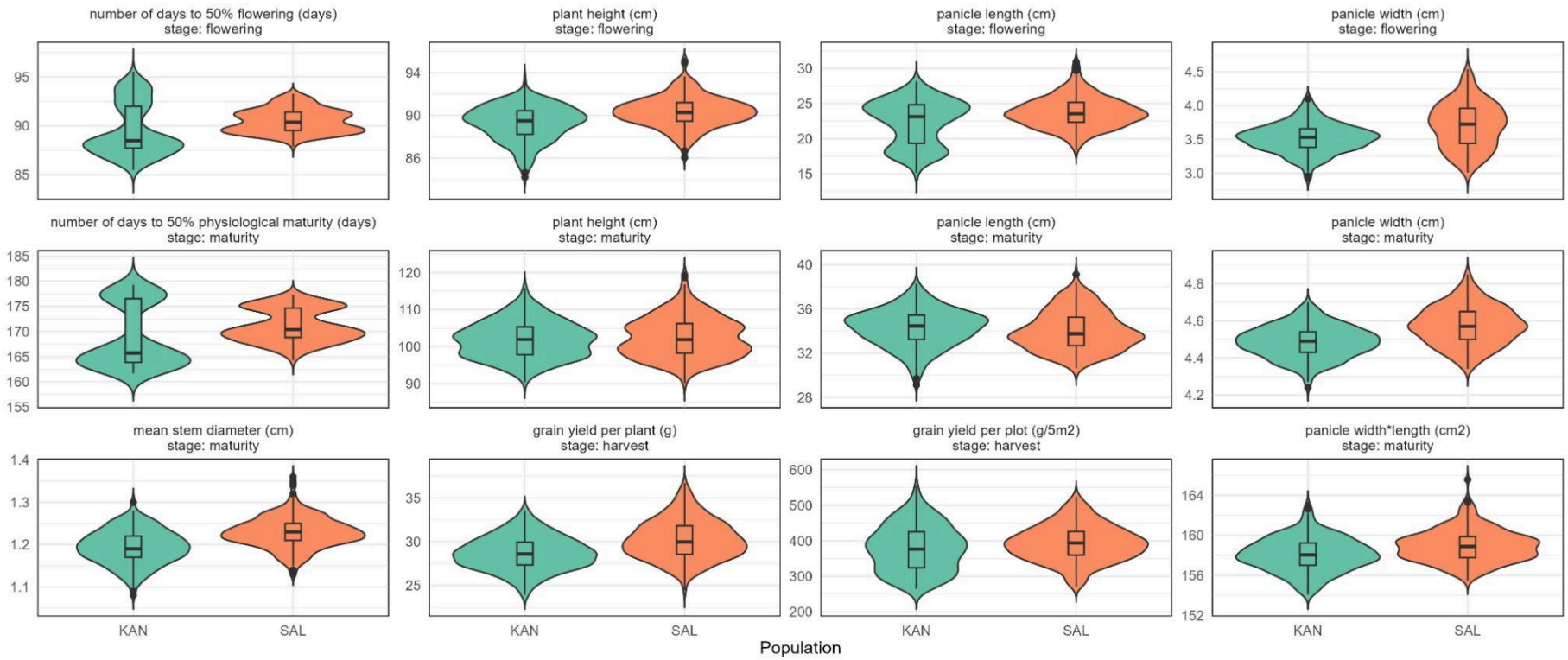
Supplementary Figure 2. Frequency distribution of sequencing depth across all SNP loci retained after variant calling.



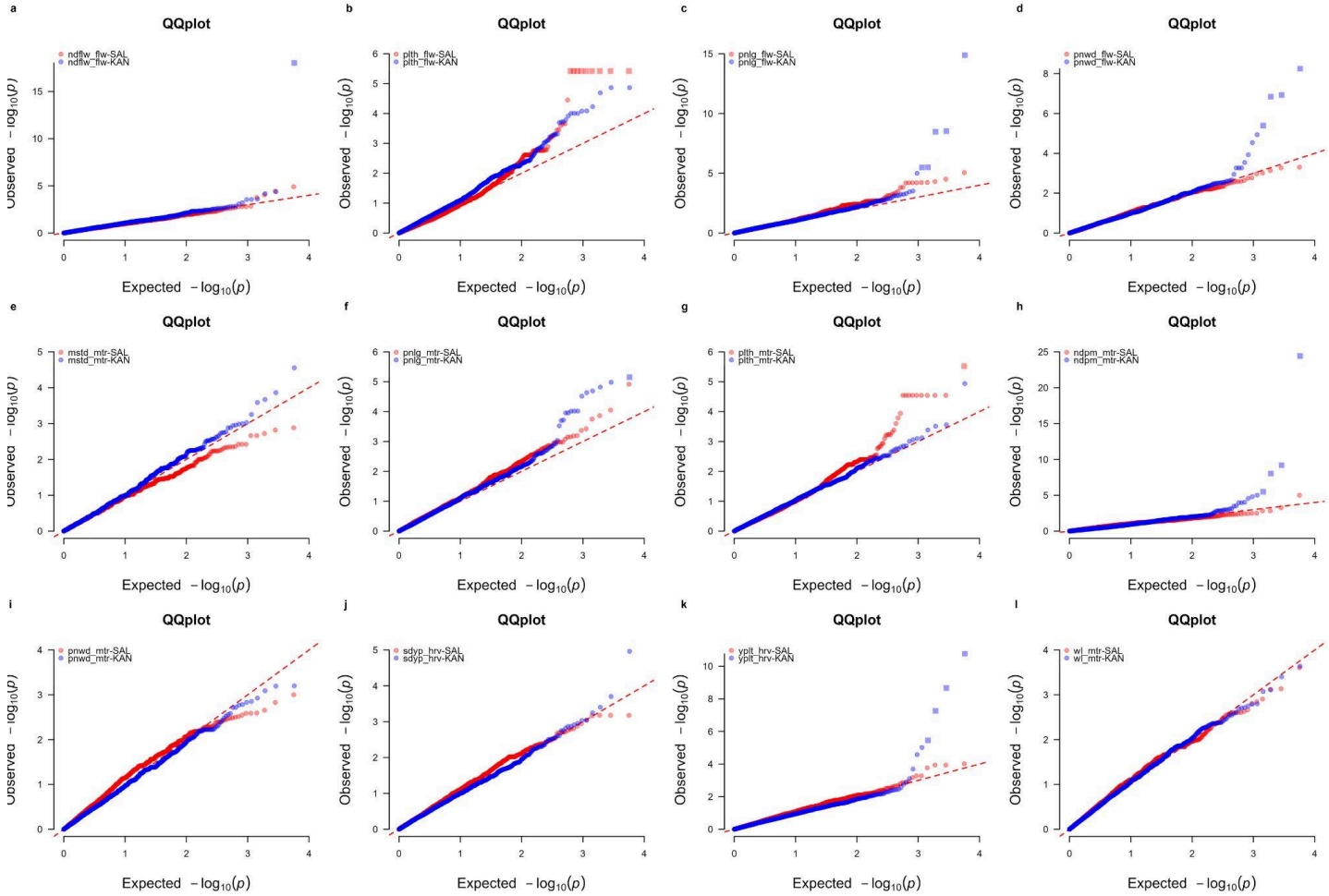
Supplementary Figure 3. Distribution of SNP markers across the quinoa genome after genotype filtering. A total of 21,258 biallelic SNPs retained after the initial BCFtools filtering step were subjected to additional filtering using the snpR R package, including a minor allele frequency threshold of 0.05 within each cross, a minimum locus call rate of 50% (min_ind = 0.50), a minimum individual genotype completeness of 50% (min_loci = 0.50), and removal of non-polymorphic loci. The final dataset comprised 5,803 SNP markers across 118 individuals.



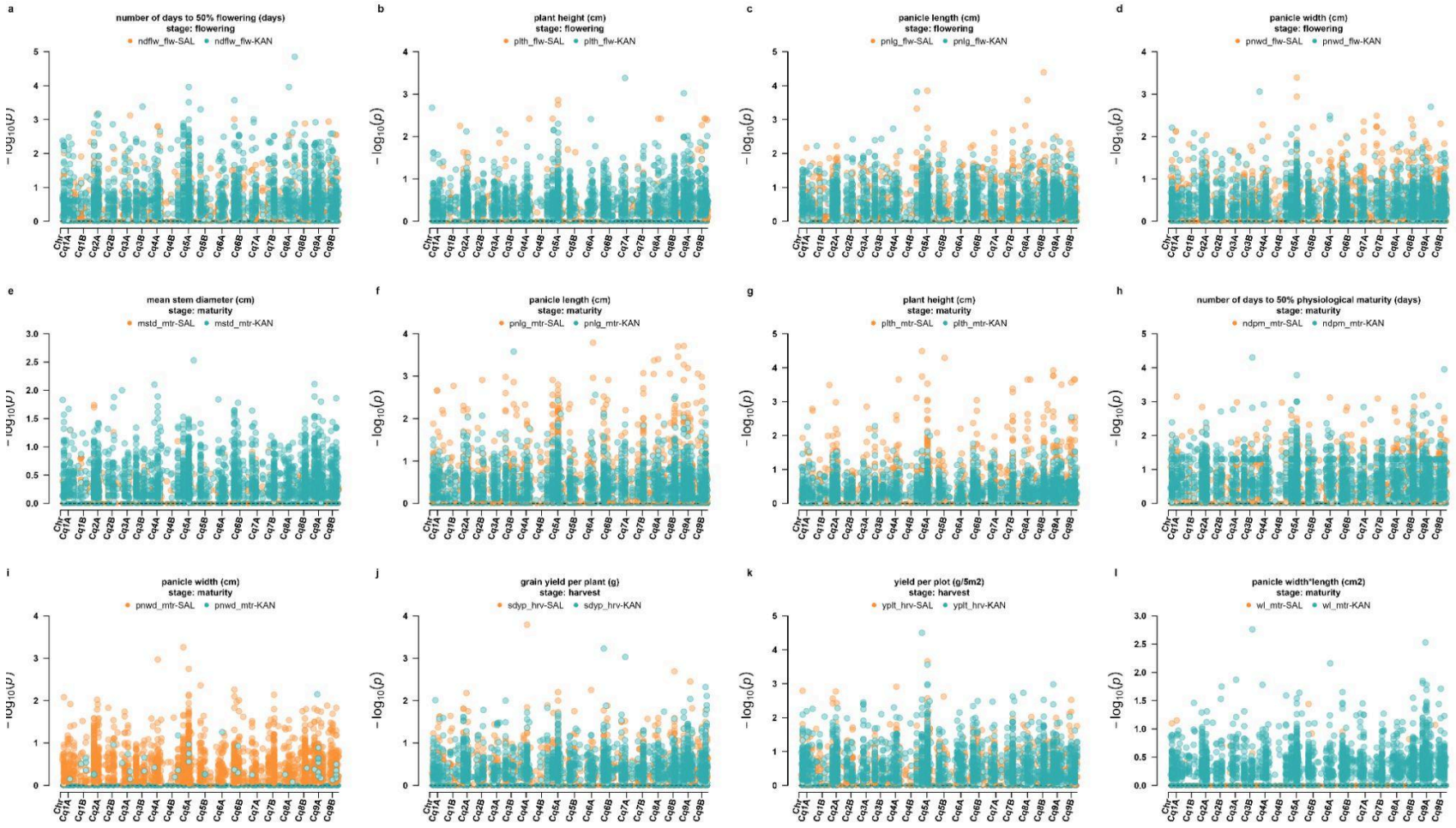
Supplementary Figure 4. Population structure inferred with sparse non-negative matrix factorization (sNMF) across six quinoa crosses. (a) Ancestry barplots for assumed ancestral clusters from K = 2–10. (b) Selection of the optimal K using the mean cross-entropy criterion shows a minimum at K = 3. Each vertical bar represents an individual, and colors indicate cluster-membership proportions; individuals are grouped by cross: Negra Collana × Kancolla (NCO×KAN), Huariponcho × Kancolla (HUA×KAN), Pasankalla × Kancolla (PAS×KAN), Salcedo INIA × Pandela Rosada (SAL×PAN), Salcedo INIA × Huariponcho (SAL×HUA), and Salcedo INIA × Negra Collana (SAL×NCO).



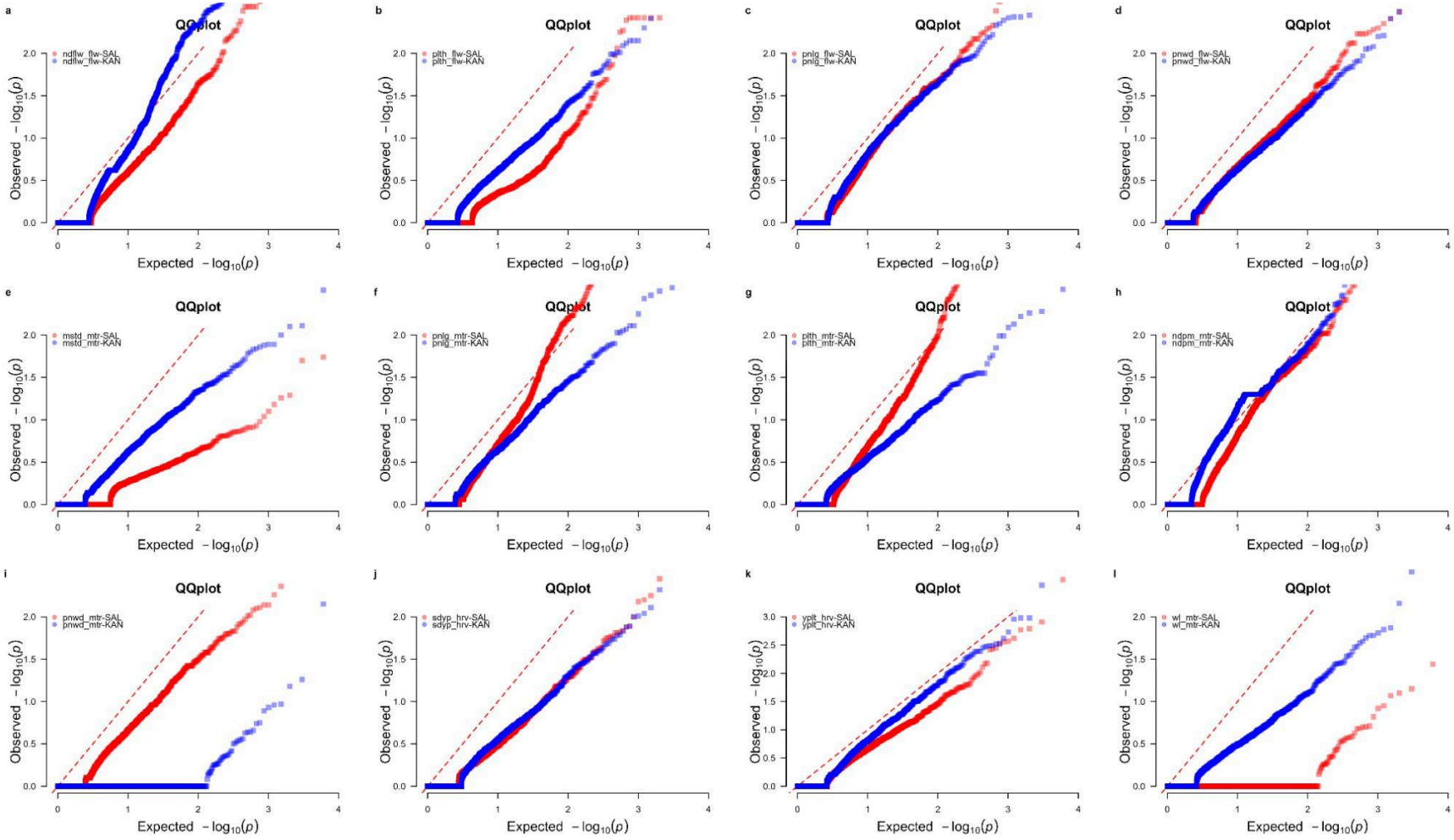
Supplementary Figure 5. Traits distribution for yield related traits in quinoa (*Chenopodium quinoa*) in two NAM populations. Kancolla (green) and Salcedo (red) NAM population under multilocation trials in the Peruvian Highlands.



Supplementary Figure 6. QQplot for Genome-Wide Association Studies in Salcedo (red) and Kancolla (blue) NAM population under multilocation trials using the *rMVP* R package (n = 118). (a) Number of days to 50% (days, ndflw), (b) Plant height (cm, plth), (c) Panicle length (cm, pnlg), (d) Panicle width (cm, pnwd), (e) Mean stem diameter (cm, mstd), (f) Number of days to 50% physiological maturity (days, ndpm), (g) Grain yield by plant (g, gypp), (h) yield per plot (g/5m2, yplt), and (i) Panicle width*length (cm2, wl).



Supplementary Figure 7. Genome-wide association study for yield related traits in quinoa (*Chenopodium quinoa*) in two NAM populations using *NAM* R package. (a) Number of days to 50% (days, ndflw), (b) Plant height (cm, plth), (c) Panicle length (cm, pnlg), (d) Panicle width (cm, pnwd), (e) Mean stem diameter (cm, mstd), (f) Number of days to 50% physiological maturity (days, ndpm), (g) Grain yield by plant (g, gypp), (h) yield per plot (g/5m2, yplt), and (i) Panicle width*length (cm2, wl). Orange and green points denote SAL- and KAN-derived NAM populations, respectively (n = 118). The horizontal line indicates the genome-wide significance at Bonferroni threshold (p-value < 8.62×10^{-6}).



Supplementary Figure 8. QQplot for Genome-Wide Association Studies in Salcedo (red) and Kancolla (blue) NAM population under multilocation trials using the *NAM* R package (n = 118). (a) Number of days to 50% (days, ndflw), (b) Plant height (cm, plth), (c) Panicle length (cm, pnlg), (d) Panicle width (cm, pnwd), (e) Mean stem diameter (cm, mstd), (f) Number of days to 50% physiological maturity (days, ndpm), (g) Grain yield by plant (g, gypp), (h) yield per plot (g/5m2, yplt), and (i) Panicle width*length (cm2, wl).